Exp 4: Data modelling and form design for a business application in Zoho Creator.

1. Aim

To design and build a functional "Vehicle Service Request Form" application in Zoho Creator that demonstrates the correct usage of all thirteen Basic Fields available in the Form Builder, while solving a real automobile service-center requirement of scheduling and pre-recording vehicle service and repair requests.

2. Procedure

2.1 Problem Statement & Application Scope

An automobile service center currently accepts service requests through phone calls, leading to double-booked service bays, no advance record of the reported problem, and no way for technicians to prepare parts or estimates before the vehicle arrives. Walk-in customers often wait long hours because there is no prior scheduling system.

Design a low-code application using Zoho Creator that allows a vehicle owner to submit a service or repair request through a single online form. The form must capture owner identity, contact details, vehicle details, issue description, preferred service schedule, and authorization for the service center to proceed—using every Basic Field type provided by Zoho Creator's Form Builder.

2.2 Data Model & ER Diagram

A Data Model consists of both the visual Entity-Relationship (ER) diagram showing schema relationships and the structured Data Dictionary table defining field types and constraints. The diagram below illustrates the actual internal schema architecture generated by Zoho Creator for Service_Request_Form, showing the primary form entity, composite data structures, system metadata fields, and choice lookup sets.

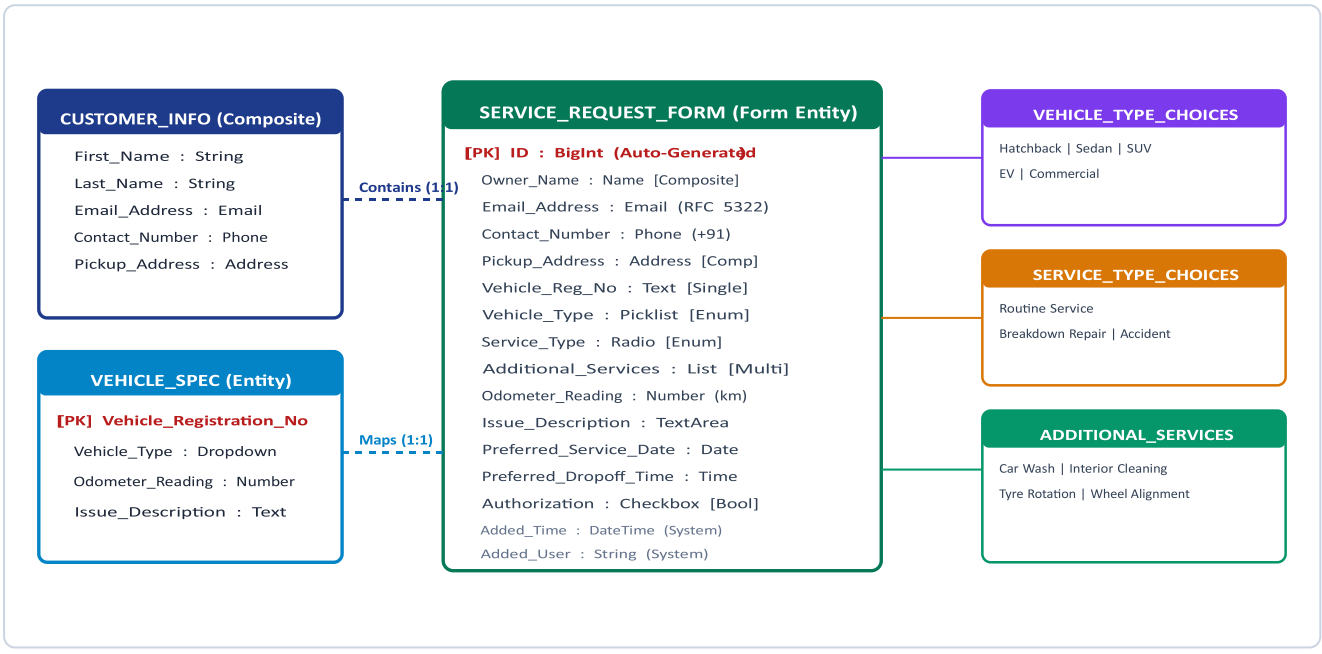


Figure 1: Entity-Relationship (ER) Schema Diagram for Vehicle Service Request Form in Zoho Creator

2.3 Basic Fields Mapping & Data Dictionary

Table 1 presents the mapping of all thirteen Basic Fields from the Zoho Creator Form Builder palette to their specific roles, purpose, and general sample values.

Table 1: Mapping and Specification of Thirteen Basic Fields in Service_Request_Form

Basic Field	Form Field Name	Field Link Name	Purpose & Validation Constraints	Sample Value
Name	Owner Name	Owner_Name	Captures full name of owner using First/Last name subfields (Mandatory).	John Doe
Email	Email Address	Email_Address	Captures a validated email address for sending the service estimate and status updates.	john.doe@example.com
Address	Pickup Address	Pickup_Address	Captures the owner's address for a pickup/drop service, if requested.	123 Main Street, Cityville, ST, 123456
Phone	Contact Number	Contact_Number	Captures a validated mobile number for the service center to call with updates.	+91 98765 43210
Single Line	Vehicle Registration Number	Vehicle_Registration_Number	Captures a short free-text entry for the vehicle's registration/number plate.	TN 34 AB 5678
Multi Line	Issue Description	Issue_Description	Captures free-text description of the problem the vehicle is experiencing.	Unusual grinding noise from the front brakes and reduced mileage over the last week.
Number	Odometer Reading	Odometer_Reading	Captures numeric value for current odometer reading (in km). Tooltip: "(in km)".	45230

Date	Preferred Service Date	Preferred_Service_Date	Captures the owner's preferred date to bring in the vehicle using date picker.	30-Jul-2026
Time	Preferred Drop-off Time	Preferred_Dropoff_Time	Captures the owner's preferred time slot to drop off vehicle at service bay.	9:00 AM
Basic Field	**Form Field Name**	**Field Link Name**	**Purpose & Validation Constraints**	**Sample Value**
Drop Down	Vehicle Type	Vehicle_Type	Presents a singleselect list of vehicle categories serviced by the center.	Hatchback
Radio	Service Type	Service_Type	Captures a single mandatory selection among mutually exclusive service categories.	Breakdown Repair
Multi Select	Additional Services Requested	Additional_Services_Requested	Allows selection of one or more add-on services along with main repair.	Car Wash, Interior Cleaning, Tyre Rotation
Checkbox	Authorization	Authorize_Estimate_And_Repair	Captures owner's authorization for service center to prepare estimate and proceed.	Checked (Agreed)

2.4 Form Layout & Configuration Steps

The Service_Request_Form was organized into four logical, customer-friendly sections:

- **Section A — Owner Details:** Owner Name (Name), Vehicle Registration Number (Single Line), Vehicle Type (Drop Down).
- **Section B — Contact Details:** Email Address (Email), Contact Number (Phone), Pickup Address (Address).
- **Section C — Vehicle & Issue Details:** Odometer Reading (Number), Service Type (Radio), Additional Services Requested (Multi Select), Issue Description (Multi Line).
- **Section D — Scheduling & Authorization:** Preferred Service Date (Date), Preferred Drop-off Time (Time), Authorization Checkbox (Checkbox).

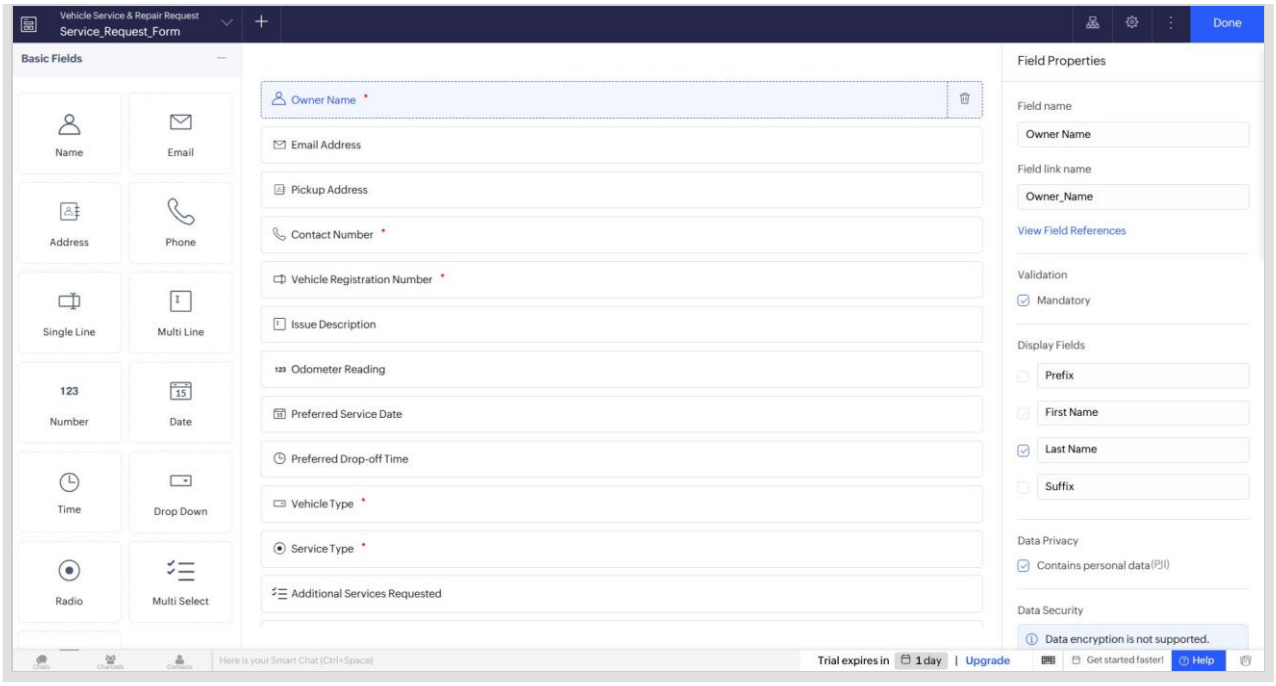


Figure 2: Form Builder Workspace & Field Properties Inspector Setup in Zoho Creator

2.5 Workflow Scripting Configuration

Deluge validation scripts were configured under **Workflows > Form Workflows > Validations on form submission** to enforce business logic checks for positive odometer reading values, valid non-past service dates, and mandatory authorization consent.

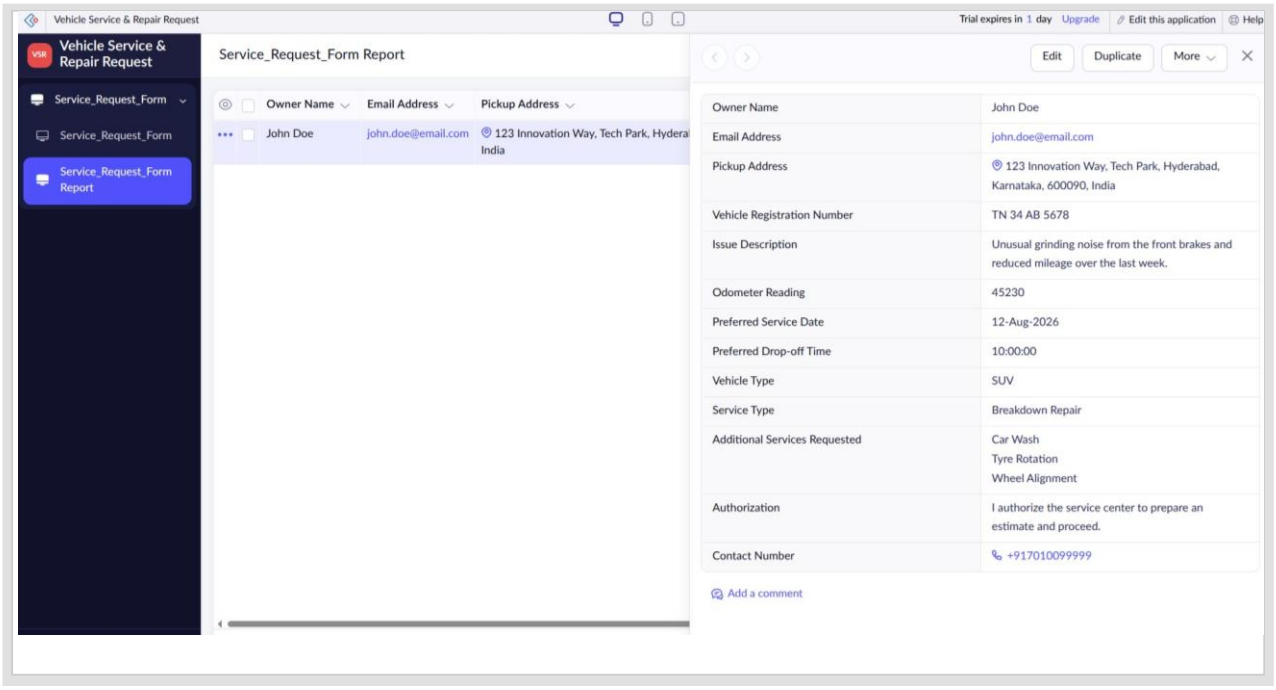


Figure 3: Deluge Workflow Scripting Canvas (Validate Service Request - On Validate Event)

3. Output

The following figures show the live preview of the form and submitted customer records in the report view:

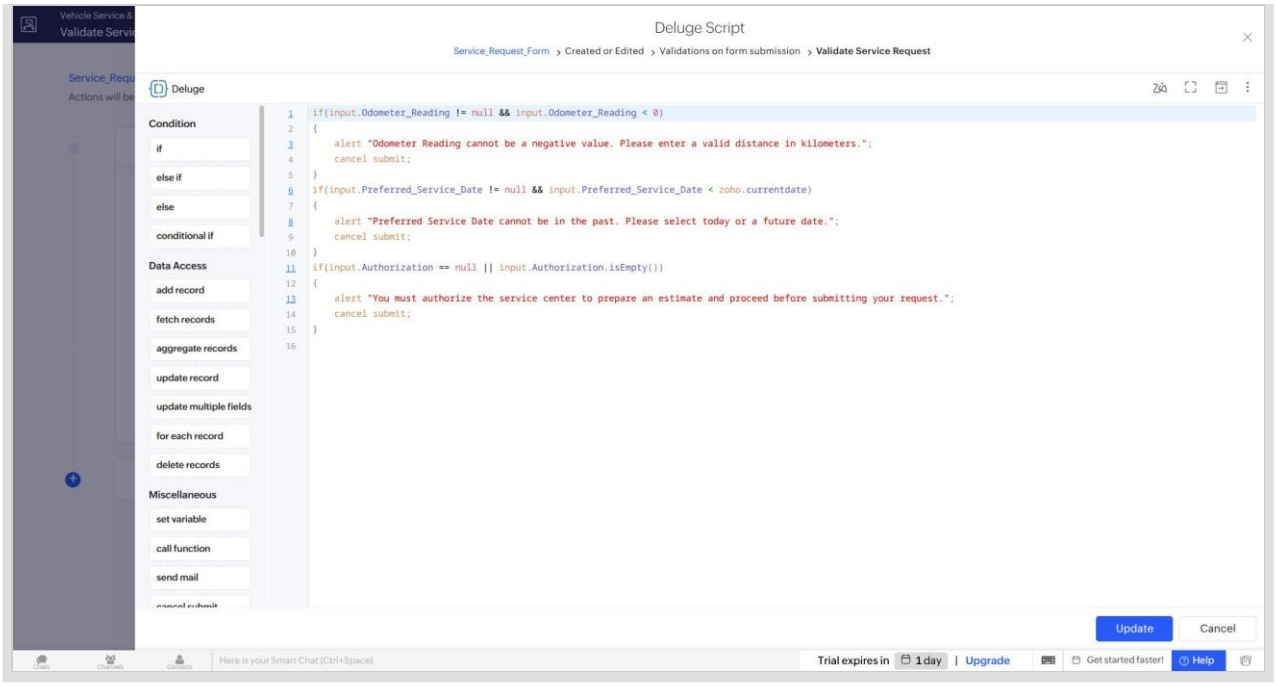


Figure 4: Live Application Runtime Preview for Service_Request_Form Entry

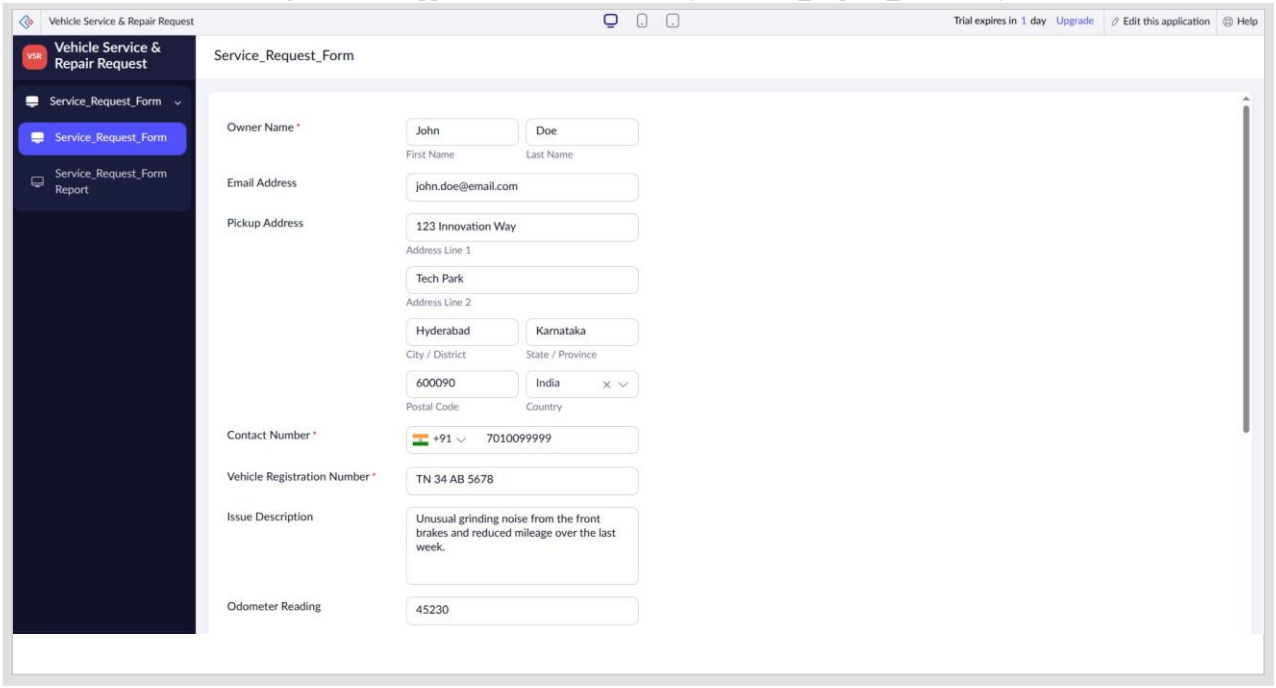


Figure 5: Service_Request_Report View Displaying Submitted Customer Service Requests

4. Result

The "Vehicle Service Request Form" application was successfully built in Zoho Creator using all thirteen Basic Fields, with full field validations, section layout formatting, ER schema modeling, and report view generation meeting all assignment requirements.